

The above substitution emphasizes the fact that we are now dealing directly with the exponential distribution of the squared magnitudes X_1 and X_2 , rather than with the complex fields Z_1 and Z_2 because σ^2 and ρ are, respectively, the variance and the correlation coefficient of X_1 and X_2 , as shown in the previous section. We now seek to extend this result to the sum of k independent correlated exponential variables, leading to correlated Gamma-distributed variables.

Following an approach analogous to that used in deriving the Nakagami distribution [1], we make use of the following two-dimensional Laplace transform formula:

$$\frac{1}{\Gamma(k)b^{k-1}} \int_0^\infty \int_0^\infty (x_1 x_2)^{\frac{k-1}{2}} e^{-\alpha(x_1+x_2)} I_{k-1}(2b\sqrt{x_1 x_2}) e^{-z_1 x_1 - z_2 x_2} dx_1 dx_2 = \frac{1}{[(z_1 + a)(z_2 + a) - b^2]^k}. \quad (20)$$

This formula, reported as equation (79) in [12], holds for $\Re(k) > 0$. By setting the parameters to match our exponential distribution, $k = 1$, $\alpha = \frac{1}{\sigma(1-\rho)}$, $b = \frac{\sqrt{\rho}}{\sigma(1-\rho)}$, we obtain the joint characteristic function of the exponential distribution:

$$\varphi_{\text{exp}}(z_1, z_2) = \frac{1}{\sigma^2(1-\rho) [(z_1 + a)(z_2 + a) - b^2]}. \quad (21)$$

Since the sum of k independent exponential variables follows a Gamma distribution, the characteristic function of two correlated Gamma variables can be directly deduced:

$$\varphi_{\text{Gamma}}(z_1, z_2) = [\varphi_{\text{exp}}(z_1, z_2)]^k = \frac{1}{\sigma^{2k}(1-\rho)^k [(z_1 + a)(z_2 + a) - b^2]^k}. \quad (22)$$

To invert this characteristic function and recover the probability density function, we use the two-dimensional Mellin inversion formula [13]:

$$\begin{aligned} \left(\frac{1}{2\pi i}\right)^2 \int_{c-i\infty}^{c+i\infty} \int_{c-i\infty}^{c+i\infty} \frac{1}{[(z_1 + a)(z_2 + a) - b^2]^k} e^{z_1 x_1 + z_2 x_2} dz_1 dz_2 \\ = \frac{1}{\Gamma(k)b^{k-1}} (x_1 x_2)^{\frac{k-1}{2}} e^{-\alpha(x_1+x_2)} I_{k-1}(2b\sqrt{x_1 x_2}). \end{aligned} \quad (23)$$

Applying this inversion formula to the characteristic function in (22), we finally obtain the joint PDF of two correlated Gamma-distributed variables:

$$f_{\text{Gamma}}(x_1, x_2) = \frac{(x_1 x_2)^{\frac{k-1}{2}}}{\Gamma(k)\sigma^{k+1}(1-\rho)\rho^{\frac{k-1}{2}}} \exp\left(-\frac{x_1 + x_2}{\sigma(1-\rho)}\right) I_{k-1}\left(\frac{2\sqrt{\rho x_1 x_2}}{\sigma(1-\rho)}\right). \quad (24)$$

This expression generalizes the previous result for the exponential PDF with $k = 1$ and agrees with results reported in the literature (see [14, 15]).

4. Probability Density Function of the Ratio of two correlated Gamma-distributed random variables

In this section, we derive the PDF of the ratio $Z = X_1/X_2$, where X_1 and X_2 are two correlated Gamma-distributed random variables with the same shape and scale parameters.

By definition, the PDF of the ratio Z for two non-negative random variables X_1 and X_2 is given by: